



BRAC UNIVERSITY

CSE461: Introduction to Robotics

Lab Project Report

Title: Autonomous SLAM-Based Search & Rescue Mapping Robot

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Abstract

Building collapses, earthquakes, fires, and industrial mishaps are examples of natural calamities that frequently create hazardous conditions for human rescue teams. By investigating dangerous areas before rescuers approach them, an autonomous search and rescue robot can enhance disaster response in Bangladesh, where flooding, fire events, and infrastructure failures are common. An ESP32-S3, microcontroller, ultrasonic sensors, MPU6050 IMU, wheel encoders, HLK-LD2410C are used in this project to create an autonomous SLAM-Based Search & Rescue Mapping Robot. The robot explores an unfamiliar area on its own, uses sensor fusion technique to create a simplified two-dimensional occupancy grid map, uses motion and heat to identify possible survivors and uses wifi to send real-time map data and alerts.

The suggested approach is appropriate for low-cost embedded hardware since it uses pseudo-SLAM rather than computationally costly SLAM methods. The robot can traverse uncharted territory while avoiding obstacles thanks to frontier-based exploration. The robot effectively created a coarse map, indicated possible survivor sites, and spotted barriers with respectable accuracy, according to experimental testing. The solution is economical for research projects and educational institutions like ours, with an estimated total implementation cost around BDT 8000. From sustainability standpoint the project reduces electronic waste and operating expenses by utilizing reusable hardware modules, energy efficient embedded components, and rechargeable batteries. The suggested solution shows how low-cost robotics and embedded technological innovation and safer disaster response in Bangladesh.

Keywords

Search and Rescue, ESP32-S3, Pseudo-SLAM, Sensor Fusion, Embedded systems.

1. Introduction

Bangladesh is vulnerable to many natural disaster like earthquakes, fires, floods, cyclones, building collapses. During such emergencies rescue teams often face dangerous environments with unstable structures, smoke, debris and limited visibility. Sending human rescuers into these environments immediately with such risks may increase the risk of injuries. Whereas sending a robot can make things easier. Intelligent rescue robot that explore dangerous regions before rescue people enter them have been made possible by recent developments in embedded systems, autonomous robotics, and sensor technologies. These robot are capable of gathering environment data, identifying potential survivors, detecting barriers, and sending information to a rescue centre.

The following are the objectives of this project:

- Create a mobile autonomous robot that can explore uncharted territories.
- Use sensor fusion to create a simpler occupancy grid map.
- Use motion and 24 GHz mmWave radar sensor to find possible survivors.
- Use wifi to provide survival notifications and mapping data.
- Create a low cost prototype that can be used for research and teaching.

The goal of this project is to develop and deploy an autonomous search and rescue robot that can avoid barriers, create a simplified occupancy grid map, explore uncharted territory, and use a 24GHz mmWave radar sensor to detect human presence. The robot performs real-time sensing

localization, and autonomous navigation by integrating ultrasonic sensors, an MPU6050 inertial measurement unit, wheel encoders, and the HLK-LD2410C mmWave radar sensor via the ESP32-S3 microprocessor. The ESP32-S3's integrated wifi-module is used to wirelessly communicate the created map and human presence data to a monitoring station. Because it shows how various sensors, motor drivers, and wireless communication modules may be integrated into a single embedded system, this project is extremely significant to the field of computer interface. This is a useful application of computer interface principles for robotics research and disaster response in Bangladesh because it offers hands-on expertise in sensor interfacing, embedded programming, real-time data collecting, motor control, sensor fusion and autonomous robotic navigation.

2. Related Work/ Inspiration

In robotics, autonomous mapping robots have been used in significant ways. Advanced LiDAR-based SLAM is used by commercial robots created by companies like Boston Dynamics and Clearpath Robotics to navigate in disaster environment. Organization like NASA, DARPA, and several robotics research labs have examined autonomous rescue robot in great detail. For precise SLAM, the majority of commercial rescue robots use pricey sensors like LiDAR, stereo cameras, and powerful onboard computers.

Ultrasonic sensors are used by a number of inexpensive educational robots to avoid obstacles. Nevertheless, autonomous mapping, survivor detection, and wireless monitoring capabilities are typically absent from these systems. The requirement for an inexpensive rescue robot is the main inspiration for this project specially for a developing country like Bangladesh.

3. Technical Approach

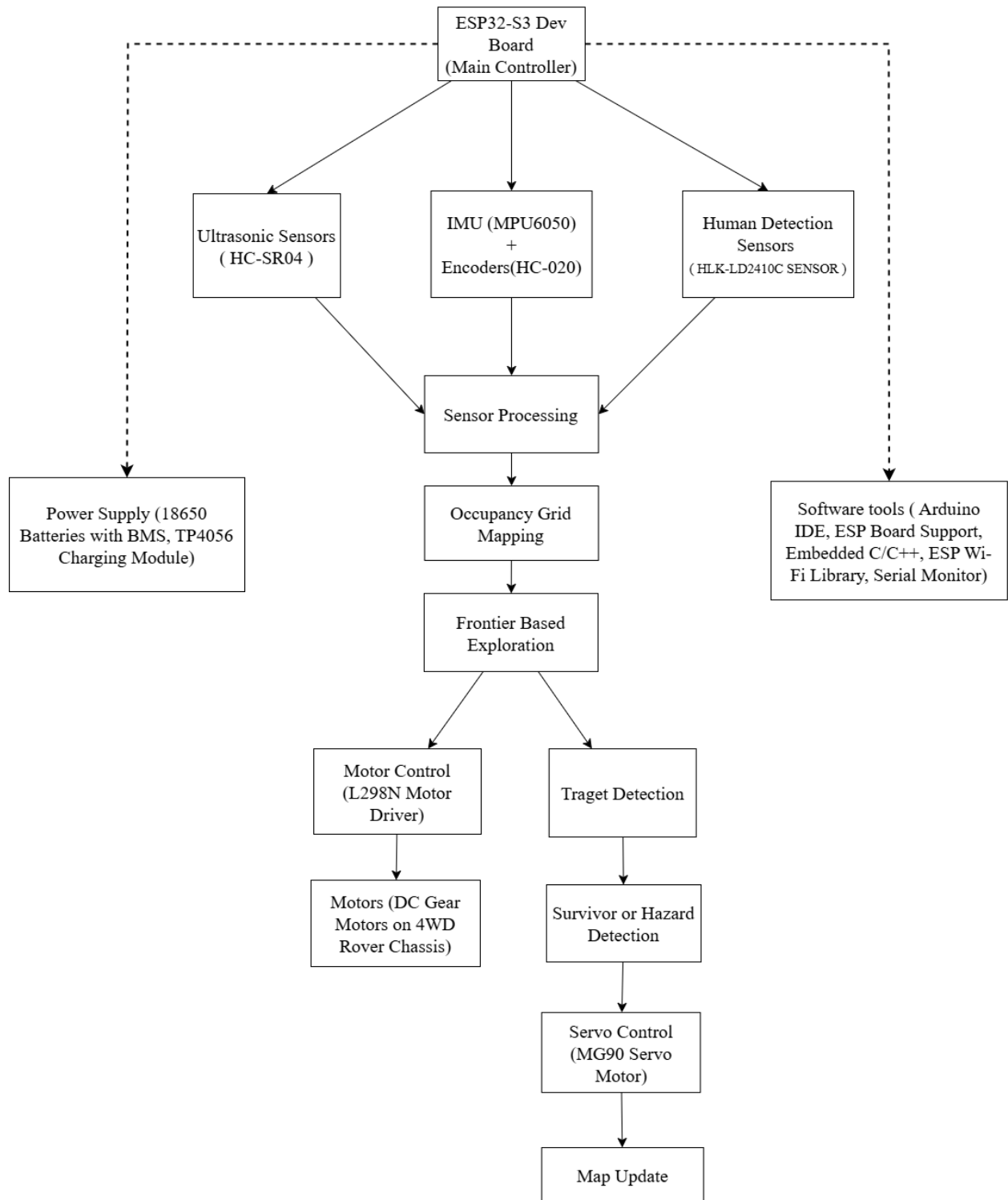


FIG: SYSTEM ARCHITECTURE

HARDWARE TOOLS:

Hardware	Description
ESP32-S3 Dev Board	This is the system's primary microcontroller. This uses built-in Wifi to transmit real-time mapping and human detection data, interpret sensor data, execute sensor fusion and autonomous navigation algorithms, and drive the motors
HC-SR04 Ultrasonic	Used to calculate the robot's distance from nearby obstacles. These sensors offer data on obstacle detection for autonomous navigation and occupancy grid mapping
MPU6050 IMU	A six-axis inertial measurement device that integrates a three axis gyroscope and accelerometer. It offers motion and orientation information to enhance localization via sensor fusion.
HLK-LD2410C SENSOR	Identify the presence of both moving and stationary people using millimeter-wave radar technology. Compared to traditional motion sensors, it allows the robot to more reliably identify possible survivors in catastrophe scenarios.
HC-020 Wheel Encoder	Location estimation is supported by measuring the wheels rotation to determine the robot's speed and distance traveled.
MG90 Metal Gear Servo Motor	Enables the robot to scan a larger area for obstacle identification and environmental mapping by rotating the ultrasonic sensor assembly.
L298N Motor Driver Module	Uses PWM signals to interface the ESP32-S3 with the DC motors, regulating

	their speed and direction
4WD Rover Chassis with DC motor gear	Provides the mobility and mechanical platform needed for independent movement across various terrains.
18650 Rechargeable Batteries with BMS	Provides the robot with portable power while the battery management system guards against short circuit ,over charging and over discharging.
TP4056 Charging Module	Offer charging protection and safely charges the rechargeable 18650 lithium-ion batteries
Jumper Wire,Breadboard,connectors,resistor	To establish electrical connections between microcontroller,sensor,motor driver and other electronics components.

SOFTWARE TOOLS:

Software	Description
Arduino IDE	For writing compiling and uploading the embedded C/C++ program to the ESP32-S3
ESP Board support (BSP)	Provide necessary libraries and drivers for programming the ESP32-S3 using arduino ide
Embedded C/C++	This is the programming language we used to implement sensor interfacing,motor control,mapping,autonomous navigation and wifi communication.

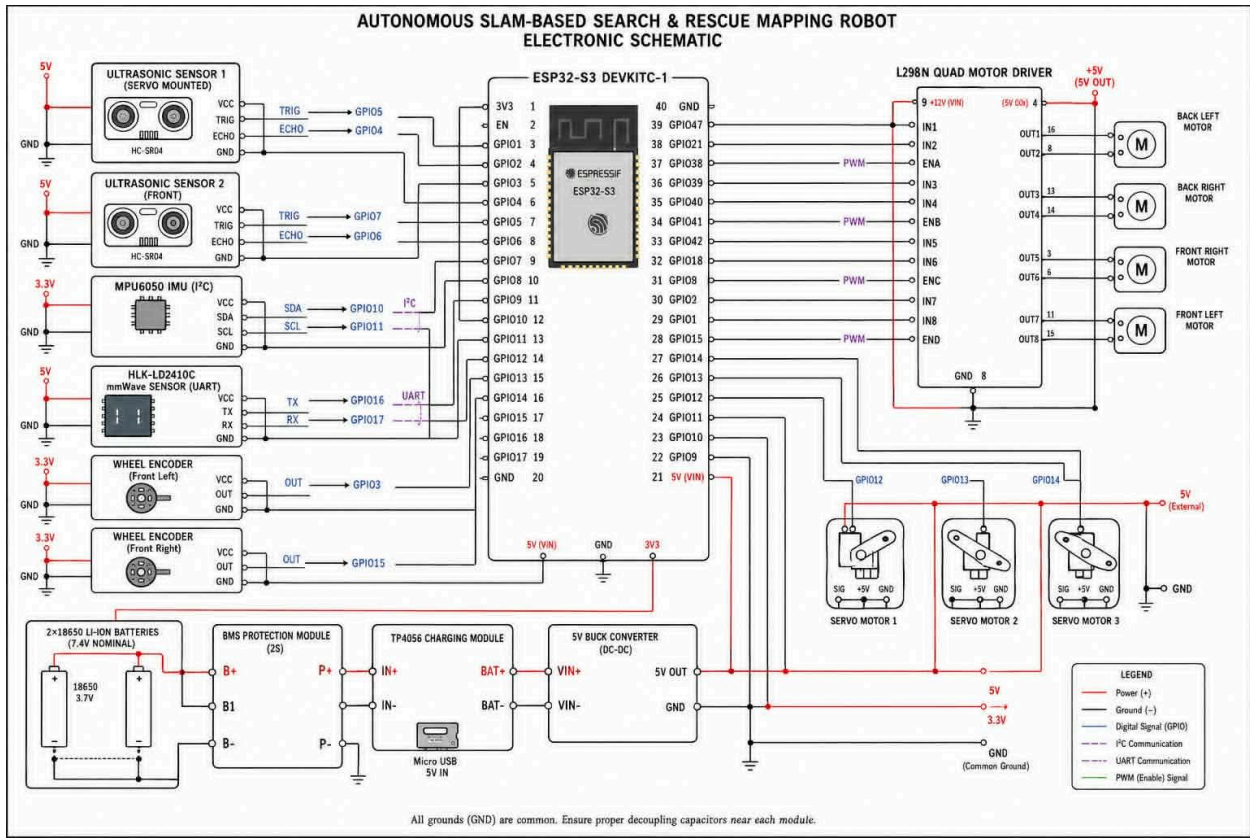
ESP WI-FI Library	Allows map data and person detection alerts to be transmitted wirelessly between the robot and the remote monitoring station.
Serial Monitor	Serial monitor is used for testing debugging, and tracking sensor reading

Cost Breakdown:

Components	Unit Cost	Quantity	Total
ESP32-S3 Dev Board	890	1	890
HC-SR04 Ultrasonic	125	4	500
MPU6050 IMU	250	1	250
HLK-LD2410C	740	1	740
Wheel Encoders	195	2	390
SG90 Servo	240	2	480
L298N Driver	895	1	895
4WD/Tank Chassis	1350	1	1350
18650 Battery + BMS	250	2	500
TP4056 Charger	350	1	350

Misc.	—	—	300
Total			7695

CIRCUIT_DIAGRAM:



Functionality

In order to explore uncharted territory,create a simplified occupancy grid map,identify human presence and return to the starting point,the autonomous robot integrates data from several sensors.Below is a description of the entire operational workflow.

Step1: Calibration and System initialization:

All connected hardware such as the ultrasonic sensors, MPU6050 IMU, wheel encoders, HLK-LD2410C mmWave human presence sensor, servo motors, and motor driver, is initialized by the ESP32-S3 when the robot is turned on. While the ultrasonic sensors assess the robot's initial distance to ensure obstacle detection, the IMU and wheel encoders are calibrated to determine the robot's initial heading and position.

Step 2: Autonomous investigation and Grid Mapping of Occupancy:

Following initializing, the robot uses a rule-based frontier exploration algorithm to start investigating the uncharted territory. While the servo-mounted ultrasonic sensor rotates from left to right to scan the surrounding environment, one ultrasonic sensor continually checks the forward direction. Using sensor fusion techniques, the gathered distance measurements are integrated with displacement information from the wheel encoders and orientation data from MPU6050. By designating each cell as free, occupied, or unknown, the ESP32-S3 updates a reduced two-dimensional occupancy grid, enabling the robot to recognize uncharted territory and safely maneuver around obstructions.

Step 3: Identifying Human Presence

The HLK-LD2410C 24 GHz mmWave radar sensor continually scans the area for any fixed or moving human presence as the robot continues to explore. The mmWave sensor offers dependable detection even while a person is motionless, in contrast to traditional PIR sensors. When the ESP32-S3 detects a human presence, it temporarily stops navigation to confirm the detection and saves the matching location on the occupancy grid.

Step 4: Transmission of Alerts and Target Marking

The robotic arm is triggered to set a color-coded physical marker close to the identified site once human presence has been confirmed, enabling rescue crews to promptly determine the location. Using its integrated Wi-Fi module, the ESP32-S3 simultaneously captures the target coordinates on the occupancy grid and sends the updated map and detection alarm to the remote monitoring station.

Step 5: Navigating Back to Home

The robot starts its return routine after the exploring procedure is over or when there is no major unexplored frontier left. The robot completes the search mission by retracing its course and returning to its starting (HOME) position using the stored odometry data that was acquired from the wheel encoders and adjusted by the MPU6050 IMU.

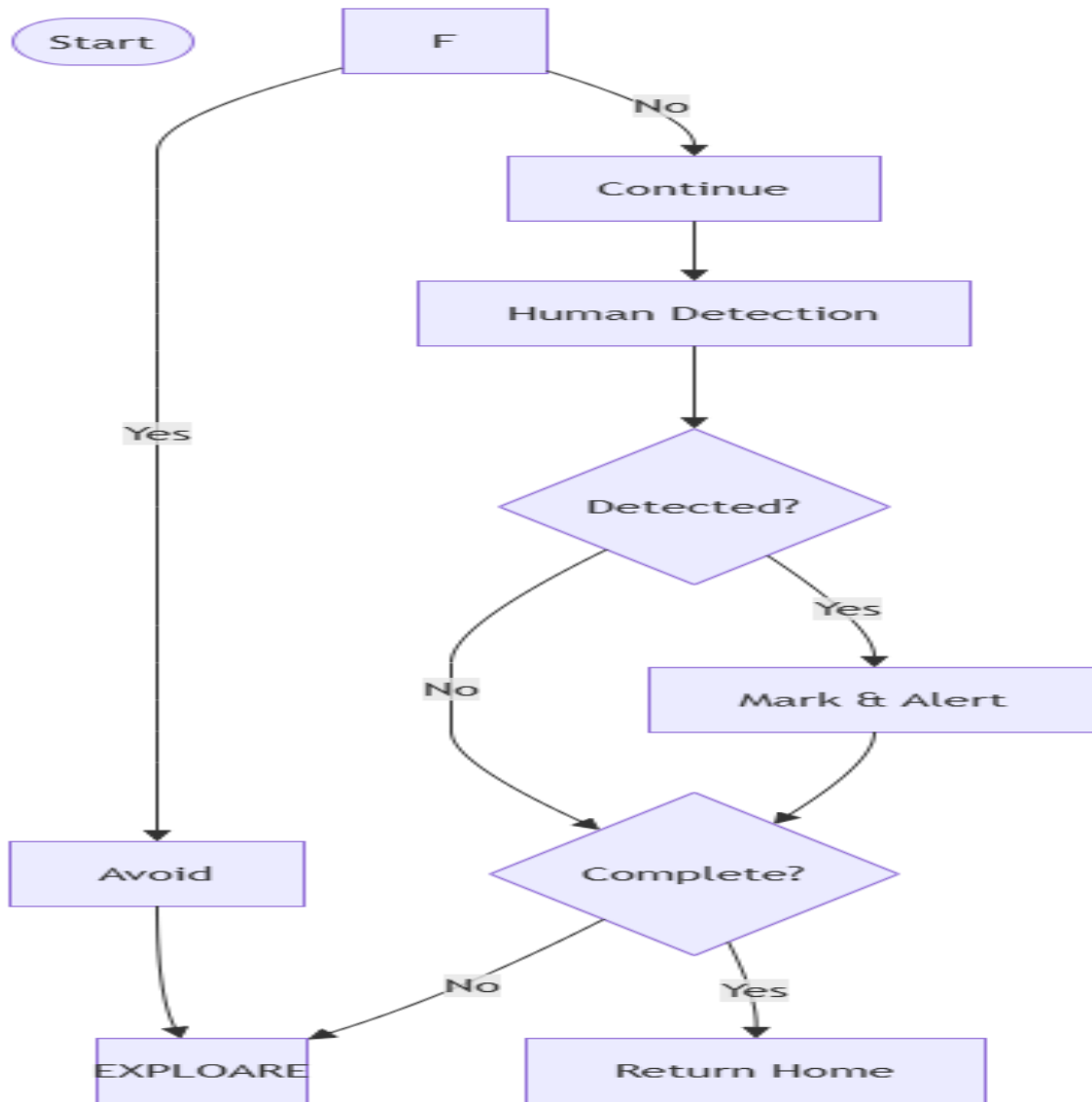


FIG: WORKFLOW DIAGRAM

CHALLENGES	LIMITATION
Servo Power Consumption and Voltage Drop: Running several servo motors at once resulted in voltage drops, which occasionally caused ESP32-S3 reset	.To give the servos a separate regulated power source while keeping a common ground with the ESP32-S3, a special DC-DC buck converter was utilized.

ESP32-S3, a special DC-DC buck converter was utilized. Ultrasonic Sensor Echo Interference: Because of overlapping echoes, continuous sweeping of the servo-mounted ultrasonic sensor resulted in erratic distance readings	To lessen echo interference and increase measurement accuracy, a step-and-wait scanning technique with a stabilizing delay of around 30 ms between servo movements was used.
False Human Detection: Occasionally, the mmWave sensor picked up unnecessary movement from nearby objects or interference from the surroundings.	In order to reduce false positives and increase detection reliability, multiple successive detections were needed before verifying human presence.

4. Sustainability & Impact

Sustainability:

The proposed autonomous search and rescue robot is designated as a reusable and modular robotic platform. The components that are used to build this includes ESO32-S3 development board, sensors, motors, motor driver, rechargeable batteries and servo motor can be reused for future robotic applications and experiments. The components can also be replaced or upgraded because of its modular structure.

The rechargeable 18650 lithium-ion batteries with a Battery Management System (BMS) provides a reusable power source . Similarly, the use of low-cost and widely available electronic components makes the system suitable for repeated laboratory testing and educational development.

Impact:

Searching and rescuing operations make humans exposed to unfamiliar, dangerous or difficult environments. However, the proposed autonomous system helps to reduce this risk by exploring the areas which are difficult and dangerous to deal with for humans. Also this robot is able to detect obstacles, construct occupancy-grid map, and locate human presence. It can also effectively communicate with a monitoring station via wireless, providing valuable information to help with quicker searches and situational awareness.

The technologies developed for this project are also useful in industrial environments for inspection of hazardous areas, navigation in warehouses, monitoring of infrastructure, inspection of equipment and indoor mapping. The system is a prototype for a laboratory setting, but its modular design offers a basis for the design of more sophisticated autonomous robots for safety, inspection, and monitoring applications.

Future Work:

This project can be further enhanced by adding more sophisticated hardware and algorithms. Further future work could involve:

- Using a more sophisticated SLAM algorithm to enhance the map and localization.
- Incremental use of LiDAR or other distance sensors for better environmental knowledge.
- Computer vision using a camera for enhanced survivability and hazard detection.
- Implementing more sophisticated sensor-fusion technology, to minimize localization errors.
- To design the mechanical system to operate on more difficult terrain.
- Developing a more evolved remote monitoring interface that will allow real time visualization of the map.
- Enlarging the battery and incorporating built-in charging technology.
- Testing the robot in bigger and more realistic environments.

Limitations:

The current system has several limitations due to its prototype-level hardware, sensing techniques, and controlled testing environment.

The drawbacks of the major ones are:

- The robot's performance is mainly assessed in a lab-made maze and not in a real disaster area.
- The mapping method is not a complete industrial grade SLAM system, but a simplified occupancy grid map.
- The surface orientation, distance and sensor interference can impact ultrasonic sensors.
- The data measured by the wheel encoder can be erroneous and make localization drift.
- Noise and drift are other problems that can occur with IMU measurements.
- In complex environments, human-presence detection with the HLK-LD2410C may not be enough to identify all the real survivors.
- The robot's battery life and processing power is finite.

- Design of mechanical and electronic components may need to be further improved to operate on rough or irregular terrain.

5. Results & Discussion

The performance of the developed robot in terms of sensing and mapping, localization, autonomous navigation, human presence recognition and wireless communication was evaluated in a controlled maze. The data from the ultrasonic sensors (HC-SR04), IMU (MPU6050), wheel encoders (HC-020), and sensor (HLK-LD2410C) were processed by the ESP32-S3 to facilitate identification of obstacles, occupancy-grid mapping, and autonomous exploration. The wheel encoders and IMU provided the information on how the robot moved and rotated, and the MG90 servo allowed the ultrasonic sensor assembly to look in different directions. Also, the program was tested using the Arduino Serial Monitor to observe how the different sensors read and how the system works, and also to determine the transfer of mapping and human-detection data through the Wi-Fi.

The performance of the generated map and the performance of the robot in navigating the map was checked by comparing the experimental results with the real physical environment. The test involved the ability of the robot to navigate the maze, to locate obstacles, to keep reasonable localization, detect human presence and return to the pre-defined HOME position. The system was also tested under uncertain sensor conditions to make sure that system would not give a wrong interpretation due to the unreliable readings. In summary, the overall testing confirmed the use of the sensing, mapping, navigation, human-presence detection and wireless communications technologies in the same robotic vehicle. Specific measurements and accuracy values that are found in the last experiments may be added into the accuracy table and figures.

The test results can be summarized as:

Test Condition	Expected Result	Actual Result	Status
No person	No Detection	No Detection	Pass
Moving person	Person Detected	Detected	Pass
Uncertain Reading	No confirmed Detection	Does not confirm as human	Pass

6. Conclusion

This project was created to make a search and rescue mapping robot that is controlled by an ESP32-S3 microcontroller along with ultrasonic sensors, an MPU6050 IMU, HC-020 wheel encoders, an HLK-LD2410C millimeter-wave radar sensor, an MG90 servo motor, an L298N motor driver, and a 4WD rover chassis. The system combines obstacle detection, occupancy-grid mapping, localization, autonomous exploration, detection of presence of a human, and communication via Wi-Fi technologies in a single robotic platform. The project illustrates the ability to use a combination of several sensors and embedded control techniques to enable a robot to acquire data from an unknown environment and make navigation decisions independently.

The project also serves as a proof of concept for the autonomous robots to aid in search and rescue and for hazardous area inspections to avoid immediate human entry into potentially hazardous environments. While there are some prototype limitations in terms of sensor accuracy, localization drift, the computational resources needed, and the controlled environment in which the prototype was tested, the prototype itself holds promise for further development. The system could be more reliable and capable of being used in bigger and more realistic environments with future improvements such as advanced SLAM, integration of LiDAR, better sensor fusion, computer vision based detection, and better mechanical design.